Local Consumer Complaint Classification: Zero-Shot, Lightweight Adaptation and Label Budgets

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Abstract

Consumer complaint routing requires accurate, reliable classification under practical label and response-time budgets. We benchmark eleven local open-weight models on a five-category Consumer Financial Protection Bureau short-narrative task and evaluate QLoRA adaptation of Gemma 4, Qwen3 and Phi-4. Each adapted model uses 500 balanced training records, 2,000 additional validation records and three training seeds, evaluated on the same 3,000 balanced test records. Balanced mean weighted-F1 gains are 3.21–9.72 percentage points, with adapted means of 0.7833–0.7857. At the same label budget, TF-IDF/SVM scores 0.7546 and validation-stopped DistilBERT averages 0.7150. With 2,000 training records, SVM reaches 0.7908; with 20,000, SVM and DistilBERT reach 0.8291 and 0.8244. Higher-budget LLM adaptation was not evaluated, leaving the equal-budget crossover unmeasured. A separate paired thinking-on/off control on 500 balanced inputs finds a Qwen3 F1 gain of 5.23 percentage points, while Gemma's 1.49-point change has a confidence interval including zero. Thinking increases mean latency substantially for both models, and adaptation does not consistently reduce it. Minority-class weaknesses persist after adaptation. Classifier choice depends on label availability and serving requirements. Findings are conditional on the preserved short-narrative corpus; temporal generalisation and concurrent serving were not evaluated.

Keywords: consumer complaint classification, large language models, QLoRA, label budgets, CFPB, local deployment

1 Introduction

The Consumer Financial Protection Bureau (CFPB) publishes complaint narratives that can support regulatory oversight and financial-service routing. Misclassification may delay a complaint or send it to the wrong department, motivating both accuracy and operational reliability [1–4]. Banks and regulators may also need to keep new complaint text on local infrastructure rather than transmit it to third-party APIs.

Supervised text classifiers require task-specific training labels; zero-shot instruction-following models instead receive the taxonomy in a prompt [5–7]. This reduces adaptation effort when taxonomies change, but introduces engineering costs in generation latency, output parsing, and hardware. A model that invents an unsupported label cannot be routed safely through a system that assumes a fixed category set.

The closest published comparison, Roumeliotis et al. [8], evaluates 14 commercial model APIs on the five-category CFPB task. Its best result is o1 at accuracy 0.777 and weighted F1 0.7764, with a reported API charge of $24.24 per thousand predictions. We examine whether local open-weight models provide competitive zero-shot performance, and how that choice changes when labelled training data are available. We evaluate balanced and reduced-imbalance samples, paired narrative-length variants, invalid outputs, and inference latency. Exact commercial test identifiers are unavailable, so cross-study comparisons remain protocol-matched rather than instance-identical.

We extend this commercial zero-shot task design with label-budget classifier controls and matched adaptation and thinking-policy comparisons. Joint analysis of prediction quality, invalid outputs and latency tests how classifier choice depends on label availability and serving requirements.

The adaptation experiment uses 500 balanced training records per model within a 16 GB GPU budget. The supervised experiments share a separate 2,000-record validation set; training-label budgets exclude these additional labels. QLoRA enables this test through low-rank updates to frozen quantised weights [9, 10]. Within-fit checkpoints are selected by validation loss on separate records.

Our contributions are:

1. Label-matched SVM and DistilBERT controls and a classical 500–20,000-record learning curve, with test-excluded splits and explicit accounting for additional validation labels.
2. Matched base/adapter evaluation of three LLMs over three training seeds, with validation-based checkpoint selection, paired uncertainty, and measured training and serving resources.
3. A paired Gemma/Qwen thinking-on/off experiment with fixed artifacts, prompts, inputs and total output-token budgets.
4. An eleven-model local zero-shot benchmark with balanced and reduced-imbalance tests, paired 250/500-character variants, invalid-output analysis and a contextual commercial-API comparison.

2 Related Work

Classical complaint classifiers use engineered text features and supervised learners. Word embeddings and contextual transformers extend these representations [5, 11]. Prior CFPB work applies topic models and transformer classifiers, while complaint prioritisation and explainability studies address operational deployment [1, 2, 12–14].

Instruction-following LLMs enable zero-shot classification alongside summarisation and response generation [6, 15, 16]. Related tourism, e-commerce, and customer-service studies compare generative and supervised models, but their taxonomies and data splits differ from this task [17–19]. Recent analyses of LLM-assisted complaint writing provide additional motivation for robustness studies on newer corpora [20, 21]; our source-described 2020–2021 corpus does not directly test that shift.

Reasoning-oriented models generate intermediate text before an answer [22–25]. Roumeliotis et al. [8] provide our closest five-category zero-shot reference. Bulgakov and Turchin [26] study a reasoning-enhanced rare-event prediction and correction framework on medical and consumer-service narratives, a different task from product classification. Güllü et al. [27] also investigate LLMs for customer complaint classification. Results from different splits and training procedures provide background rather than instance-matched performance targets.

Low-rank adaptation freezes base parameters and learns small task-specific updates [9]; QLoRA combines these updates with NF4, double quantisation and paged optimisation [10]. We use these established methods to test the practical value of limited complaint labels on consumer hardware. Our contribution is an empirical local deployment comparison across label budgets and output-reliability requirements, with preserved test sets for supervised baselines. The extended background discussion is supplied in the supplementary material.

3 Dataset and Preprocessing

3.1 Data Source

We use the CFPB Consumer Complaints dataset originally published by Tiwari [28] on Kaggle, which is a processed subset of the public CFPB Consumer Complaint Database. The source study describes the coverage as March 2020 to March 2021 [8]; the local processed CSV does not retain submission dates or complaint identifiers to independently verify this interval. It contains three columns: an unnamed index, `product` (the category label), and `narrative` (the free-text complaint). The raw file comprises 162,421 records consolidated to five target product categories: `retail_banking`, `credit_reporting`, `credit_card`, `mortgages_and_loans`, and `debt_collection`.

This is the same dataset and label set used by Roumeliotis et al. [8]; exact commercial test identifiers are unavailable, so comparability is at the task-design level: shared source, five-category taxonomy, balanced sample size and JSON-output instructions. Identity of the complete normalisation, length-filtering and deduplication pipeline with the reference implementation has not been independently established.

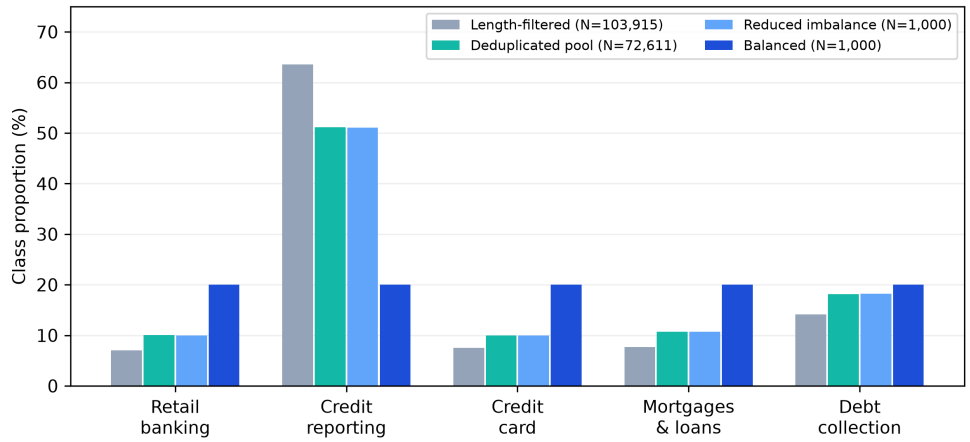


Fig. 1 Class proportions (%) in the length-filtered dataset before deduplication (N=103,915), the deduplicated sampling pool (N=72,611), the reduced-imbalance experimental subset (N=1,000), and the balanced subset (N=1,000). Experiment B rounds the proportions of the deduplicated N=72,611 pool, where credit reporting accounts for 51.15 %; the first bars describe the earlier preprocessing stage.

3.2 Class Distribution

The non-missing corpus contains 162,411 records, of which 56.1 % are credit-reporting complaints. Filtering out narratives longer than 500 characters leaves 103,915 records and raises this share to 63.6 %. Exact narrative deduplication removes 31,304 records, leaving a 72,611-record sampling pool with 51.15 % credit reporting, 18.17 % debt collection, 10.70 % mortgages/loans, 10.02 % retail banking and 9.97 % credit cards. Fig. 1 distinguishes these preprocessing stages from the experimental samples.

3.3 Preprocessing Pipeline

The preprocessing steps, implemented in Python using pandas, are applied in the following order to every record:

1. Drop rows with empty `product` or `narrative`.
2. Retain only records belonging to the five target categories.
3. Remove narratives longer than 500 characters, reducing processing time and GPU memory pressure while avoiding truncation artifacts during inference.
4. Remove exact duplicate narratives before normalisation.
5. Standardise narrative text: convert to lowercase, remove special characters, and normalise whitespace.

3.4 Experimental Subsets

Two experimental conditions are defined, each evaluated under a three-subset, two-length design to describe sampling variation.

Experiment A (Balanced): Each subset randomly samples 200 records per category ($N = 1{,}000$), following the balanced design of Roumeliotis et al. [8].

Experiment B (Reduced imbalance): Each $N = 1{,}000$ subset rounds the deduplicated-pool proportions to 51.1 % credit reporting, 18.2 % debt collection, 10.7 % mortgages/loans and 10.0 % each for retail banking and credit cards. A majority-only classifier would achieve 51.1 % accuracy. This condition measures sensitivity to class proportions within the processed pool, rather than representing deployment traffic.

Test-subset and narrative-length design: Each condition is replicated with three subset identifiers (1, 2, and 3) and at two narrative character-length limits (250 and 500 characters), yielding 12 datasets per model and 132 inference runs in total (11 models × 3 test subsets × 2 conditions × 2 length limits). The 250-character subsets contain exactly the same records in the same order as their 500-character counterparts, with narratives truncated to 250 characters. Balanced and reduced-imbalance subsets share 2,067 of their 3,000 records per condition at 500 characters, yielding 3,933 unique full-text inputs across both conditions. They are therefore not independent replications.

The eleven-model GGUF benchmark tables use the 500-character variant and mean ± standard deviation across the three test subsets. HF adaptation comparisons instead pool 3,000 predictions per mode and condition, with paired cluster-bootstrap intervals. The three test identifiers are distinct from the three LLM adaptation training seeds. The 250-character runs provide a narrative-length sensitivity check discussed in Section 5.8. These subsets describe variation within this short-narrative population; they do not establish stability across training seeds, prompts or future data.

All evaluation protocols use subsets of the same fixed benchmark. Model selection for adaptation considers zero-shot performance and deployment characteristics (Section 4.6); checkpoint and hyperparameter selection use the separate validation set (Section 4.5). Supplementary Fig. S1 describes the source narrative-length distribution.

4 Experimental Setup

The three LLM experimental arms answer distinct questions. The eleven-model GGUF benchmark evaluates zero-shot deployment configurations; paired HF runs estimate adaptation gains against matched frozen bases; the GGUF thinking control estimates policy differences on a fixed 500-input subset. Absolute scores across these arms are not directly interchangeable because artifacts, inference settings and evaluation samples differ. Adaptation and thinking gains are therefore interpreted within their respective paired controls.

4.1 Infrastructure

The supervised classifier and paired HF adaptation experiments run on an Ubuntu workstation with an Intel Core i7-14700 processor, 32 GB RAM, and an NVIDIA GeForce RTX 5060 Ti with 16 GB VRAM. The adaptation environment uses Python 3.12, PyTorch 2.14.1, Transformers 5.18.0, PEFT 0.21.2 and bitsandbytes 0.50.2. Model revisions, dataset and adapter SHA-256 hashes, package versions and run statuses are saved. The HF evaluation uses cached weights in offline mode; complaint narratives are not sent to a third-party model service.

The GGUF zero-shot benchmark uses Ollama [29], a self-contained serving runtime that manages model loading, KV-cache management, and the local REST API. Its

Table 1 Open-weight model builds evaluated in the GGUF benchmark. Reported artifact sizes are listed; peak runtime VRAM was not recorded. Type records the original grouping, not a verified per-request thinking flag.

Model	Family	Type	Params	Size
`gemma2:9b`	Google	Standard	9B	5.4 GB
`gemma4:12b`	Google	Unverified	12B	7.6 GB
`mistral:7b`	Mistral	Standard	7B	4.4 GB
`mistral-nemo:12b`	Mistral	Standard	12B	7.1 GB
`llama3.1:8b`	Meta	Standard	8B	4.9 GB
`phi4:14b`	Microsoft	Standard	14B	9.1 GB
`deepseek-r1:8b`	DeepSeek	Reasoning	8B	5.2 GB
`deepseek-r1:14b`	DeepSeek	Reasoning	14B	9.0 GB
`qwen3:8b`	Alibaba	Reasoning	8B	5.2 GB
`qwen3:14b`	Alibaba	Reasoning	14B	9.3 GB
`gpt-oss:20b`	OpenAI	Reasoning	20B	13.0 GB

Python inference script communicates through a local REST endpoint. GGUF request timing and HF generation timing use different runtime settings and measurement boundaries and are reported separately.

4.2 Models

Table 1 lists the eleven Ollama tags and reported artifact sizes. They were evaluated without task-specific fine-tuning. The families include LLaMA [30], Mistral [31], Gemma 2/4 [32, 33], Phi [34], Qwen [25], DeepSeek [23, 24] and OpenAI's open-weight gpt-oss [35]; the latter uses local weights, without a proprietary API.

Model tags do not isolate architecture, parameter count or reasoning. Gemma 4 supports configurable thinking [33], but its mode was not recorded in the eleven-model benchmark. Its leading zero-shot point estimate therefore cannot be attributed to thinking being enabled or disabled. DeepSeek's current 8B/14B tags map to R1-0528-Qwen3 and R1-Distill-Qwen2.5, respectively [36]; missing artifact digests prevent a clean size comparison. Current metadata also shows differing quantisation formats, including Q4_K_M for Qwen3 8B and MXFP4 for gpt-oss. Registry tags alone do not uniquely identify the evaluated weights. Artifact sizes are not peak runtime VRAM measurements. The HF adaptation and GGUF thinking controls pin artifacts and report their protocols separately.

4.3 Prompt Design

A common task instruction and five-label taxonomy are used across the LLM experiments, with arm-specific chat templates, thinking policies and generation limits rather than identical rendered model inputs. The instruction follows the JSON-output template used by Roumeliotis et al. [8] to align task instructions:

You are an AI assistant specializing in consumer complaint classification. Your task is to analyze a consumer complaint and classify it into the most appropriate category from the predefined list: ["retail_banking", "credit_reporting", "credit_card", "mortgages_and_loans", "debt_collection"]. Provide your final classification in the following JSON format without explanations: `{"product": "chosen_category_name"}`*. Complaint: [NARRATIVE]*

For the eleven-model GGUF benchmark, temperature is set to 0 to reduce sampling variability and improve reproducibility. No explicit `top_p`, `max_tokens`, stop-sequence, or model-specific thinking-mode flags are set; Ollama defaults are used for all other decoding parameters. Although the prompt asks for JSON without explanations, some reasoning-oriented builds may still emit intermediate text before the final answer. The inference script therefore extracts the final JSON object from the response by searching for the last valid `{"product": "..."}` occurrence using a regular expression, accounting for models that prepend reasoning traces before the answer.

The GGUF benchmark script retries an HTTP failure once and counts unresolved errors. It saves at most 2,000 response characters, replaces newlines with spaces and falls back to a category mentioned in prose if JSON extraction fails. Separate thinking text, termination reasons and token counts were not retained. Section 5.11 checks whether strict JSON reparsing changes the classifications.

4.4 Evaluation Metrics

Each GGUF benchmark run scores all $N = 1{,}000$ predictions using accuracy, weighted precision, recall and F1. Macro-F1 gives equal weight to the five classes and complements weighted metrics under imbalance. Invalid outputs include parsing failures and out-of-taxonomy labels; they remain errors in the full denominator. Mean request latency and invalid-output counts accompany the scores. All local runs incur zero model-API fees; local compute and electricity costs are not estimated.

4.5 Supervised Data and Classifier Controls

To assess whether an LLM is warranted when labelled data are available, we add TF-IDF/LinearSVC and fine-tuned DistilBERT [37] baselines. All twelve saved test sets remain unchanged. Training and validation use separate balanced sets of 20,000 and 2,000 records, respectively. We exclude normalised exact matches and shared 250-character prefixes with any test narrative, discard 133 conflicting-label text groups, and screen near duplicates using 128-permutation MinHash retrieval followed by exact word-trigram Jaccard similarity at threshold 0.85. This excludes 496 additional test-similar records and 28 validation-similar training candidates. The approximate retrieval cannot guarantee removal of every paraphrase.

The preserved test narratives are reserved before drawing training and validation records. Test-text matching is an exclusion constraint during split construction, not a source of fitted features. Verification finds zero normalised exact or shared 250-character-prefix matches between training, validation and tests at all four training budgets. Identity checks use SHA-256 hashes of normalised text and its 250-character prefix; the saved source index is a local CSV index, not a CFPB complaint identifier. Split hashes are recorded in a manifest; test-file hashes are rechecked before paired

HF evaluation. Supplementary Section 2 details the split counts, sampling seeds and statistical units.

Nested balanced training subsets of 500, 2,000, 10,000, and 20,000 records are used for the classical learning curve. TF-IDF uses word unigrams/bigrams, sublinear term frequencies, minimum document frequency 2, and at most 100,000 features. Vocabulary and IDF weights are fitted exclusively on each training subset; validation and test inputs are transformed without refitting. This differs from the fixed, non-learned lowercasing and character/whitespace normalisation. LinearSVC's C is selected from $\{0.1, 1, 10\}$ by validation weighted F1. Warm serial CPU latency includes vectorisation and prediction.

The encoder is `distilbert/distilbert-base-uncased`, pinned to the revision listed in the supplementary material. The 20,000-record and fixed 500-record controls train for three epochs with learning rate 2×10^{-5}, batch size 16, weight decay 0.01, a 256-token cap, and 10 % warmup. The checkpoint with highest validation weighted F1 is selected. Training seeds are distinct from the three saved test-subset identifiers. Within each supervised fit, checkpoint and hyperparameter selection use the separate validation set, not test performance. The training and prediction artifacts report exact model commits, training time, peak allocated VRAM, and warm serial GPU latency.

We additionally train DistilBERT on the identical 500-record subset with three training seeds and the same 2,000-record validation set. A fixed three-epoch control had a validation trajectory that was still improving, motivating a separate convergence control before inspecting its test scores. This control allows at most 30 epochs, uses patience 5 on validation weighted F1, and selects the best validation checkpoint. All other encoder settings are retained; original three-epoch outputs are preserved separately. This matches label budgets, not training-update or compute budgets.

All supervised budgets use an additional 2,000 validation labels: total label counts are 2,500, 4,000, 12,000 and 22,000 for the classical training budgets. Each LLM adapter fit likewise uses 2,500 labels in total. A training budget is not the total annotation requirement. Matching label availability does not equate validation objectives, fit counts or compute effort (Supplementary Section 2).

4.6 Controlled Lightweight LLM Adaptation

We select Gemma 4 12B, Qwen3 14B and Phi-4 14B to represent the leading zero-shot GGUF model, a configurable-thinking model and a fast generative classifier. This exploratory choice uses benchmark outcomes and deployment characteristics; adaptation findings are conditional on these selected models, rather than an independent model-selection test. The HF artifacts are `google/gemma-4-12B-it`, `Qwen/Qwen3-14B` and `microsoft/phi-4`. Their exact revisions are listed in the supplementary material. Adaptation gains use matched HF base predictions from the same pinned weights and inference settings.

Each model uses the same balanced 500-record training subset (100 per class) and the common 2,000-record validation set (400 per class), prepared with the test-exclusion procedure in Section 4.5. Reported training budgets exclude the additional validation labels. We use LoRA [9] with rank 8, scaling factor 16 and dropout 0.05, applied to attention and feed-forward projections. Gemma and Qwen use separate

query/key/value, output, gate/up/down projections; Phi uses fused QKV and gate/up projections plus output/down projections. QLoRA [10] freezes NF4 double-quantised linear weights with BF16 compute. Unquantised matrices remain BF16 and one-dimensional normalisation parameters FP32. Only adapter weights are updated.

Training uses seeds 1, 2 and 3 on the same training and validation records, each with three epochs, batch size 1, gradient accumulation 16, learning rate 2×10^{-4}, paged 8-bit AdamW, cosine decay, three warmup steps, gradient checkpointing and a 512-token cap. Loss is restricted to the assistant's target JSON and end-of-turn token; the prompt is masked. All training and validation sequences fit without truncating a target. Each fit completes 96 optimiser steps. Minimum completion-only validation loss, evaluated after each epoch, selects and records a checkpoint separately for each fit. Test scores do not guide selection.

Paired inference loads one NF4 PEFT model and enables/disables its unmerged adapter. Both modes retain identical frozen weights, normalisation, chat template and precision. The same five-category prompt is rendered through each pinned model's chat template with thinking disabled. Greedy decoding uses a 64-new-token cap and the model's end-of-turn token, with independent KV caches for every call. Inputs are left-padded with attention masks. Maximum batch size is 32. Phi's seed-1 evaluation falls back to 16 after a long-input memory failure; seeds 2 and 3 retain a maximum of 32. Both modes of a pair use the same actual batch size. Full raw responses, token counts, output-limit flags, source hashes and original row indices are saved. A JSON decoder accepts the last explicit string-valued `product` field; parse failures and out-of-taxonomy labels remain errors.

The full design yields 216,000 predictions: three models, three training seeds, two modes and twelve preserved 1,000-record files. Metrics pool the three test subsets within each condition/length.

We compute unadjusted 95 % percentile intervals from 5,000 paired bootstrap resamples using identical normalised-text-cluster weights for both modes, stratified by class when cluster labels are consistent. Exact two-sided McNemar tests compare accuracy on unique source inputs, retaining the first subset/index for duplicates; conflicting-label inputs would be excluded only from this test. Holm correction covers all 36 model–training-seed–condition–length comparisons; these are paired tests within one design, not 36 independent experimental replications. Supplementary Section 2 defines the text clusters and conflicting-label handling.

Metrics are calculated per fit, then mean and sample standard deviation are reported over the three fits. No independence between overlapping conditions or length variants is assumed. Bootstrap confidence intervals describe test-output uncertainty within a fit; training-seed variability is reported separately. Repeated model fits are never pooled into 9,000 independent test observations. The frozen base is re-evaluated in every paired run. Its small variation for Phi is run/batch variation rather than variation in trained base weights; between-fit adapter dispersion is conditional on these runs.

Training/validation runtime excludes initial weight loading; peak allocated CUDA memory includes the loaded model and training operations. Serving latency is measured separately on the first 20 records per class in the first balanced 500-character subset (100 per mode/model), sorted back to their original indices, after a warmup call in

Table 2 Supervised balanced-test performance. F1 is computed on the pooled 3,000 records; DistilBERT values average three separate model fits. Every row uses 2,000 additional validation labels; total labelled budgets are training records plus 2,000. Latency is warm serial end-to-end feature/token preparation and inference.

Model	Training records	Weighted F1	Latency (ms)
TF-IDF/SVM (CPU)	500	0.7546	0.29
TF-IDF/SVM (CPU)	2,000	0.7908	0.29
TF-IDF/SVM (CPU)	10,000	0.8241	0.28
TF-IDF/SVM (CPU)	20,000	0.8291	0.29
DistilBERT (3 epochs)	500	0.5329 ± 0.1522	2.06
DistilBERT (early stop)	500	0.7150 ± 0.0208	2.07
DistilBERT (GPU)	20,000	0.8244 ± 0.0046	2.07

each mode. Batch-1 timing includes prompt rendering, tokenisation and synchronised CUDA generation, excluding output decoding and network overhead. Full-test batch wall time divided by batch size is saved as amortised time and is not interpreted as serial latency.

5 Results

We first report direct classifiers at explicit label budgets, followed by matched HF adaptation and controlled GGUF thinking policies. The eleven-model zero-shot benchmark then supplies context on model configurations, class imbalance and published API results.

5.1 Supervised Baselines on the Preserved Test Sets

The classifier controls use the splits and settings in Section 4.5. Table 2 summarises the learning curve and the DistilBERT results at both training-label budgets. The 20,000-record encoder fits take approximately 80 s each and peak at 1.30 GiB allocated memory; their warm serial inference averages 2.07 ms on the same workstation used for the HF adaptation benchmark.

Paired two-sided McNemar comparisons against the Gemma 4 GGUF zero-shot balanced predictions (Table 6) give Holm-adjusted $p < 0.001$ for the 20,000-record SVM and for each of the three 20,000-record DistilBERT fits, with positive accuracy differences of 4.67–5.60 percentage points for these encoder fits. Correction covers all 42 baseline–reference–condition comparisons. This tests accuracy gains, not F1 differences; the zero-shot and supervised label budgets differ.

Figure 2 shows the observed training-budget trade-off. At 500 training records, adapter mean F1 exceeds the label-matched SVM point estimate. At 2,000 training records, SVM reaches 0.7908, above the three observed 500-record adapter means, with warm CPU latency below 0.3 ms. This latter comparison uses different total label budgets (4,000 versus 2,500). The LLM learning curve and a crossover budget remain unmeasured; these observations do not establish a threshold beyond which adaptation is unnecessary.

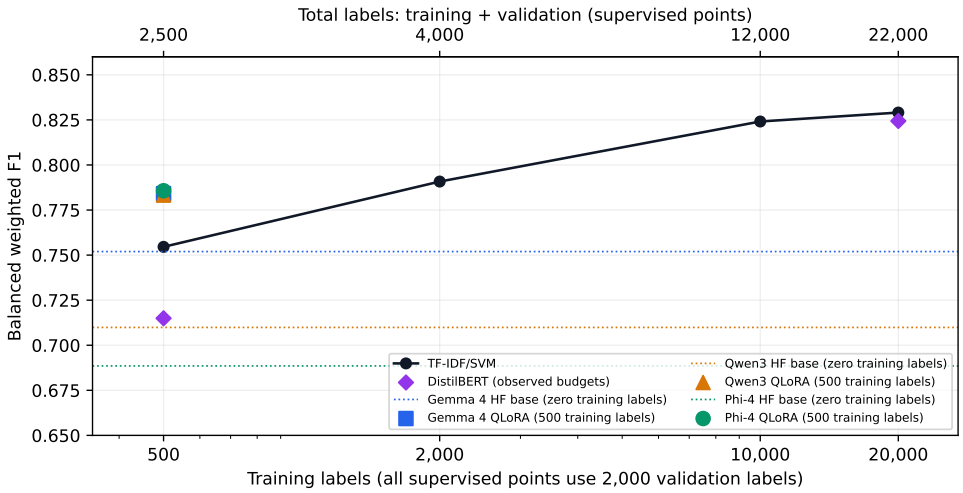


Fig. 2 Observed balanced 500-character weighted F1 by training-label budget (log axis). Direct classifiers additionally use 2,000 validation labels; the upper axis shows their total annotation requirement. QLoRA points use 500 training plus 2,000 validation labels. DistilBERT and QLoRA points are means over three fits; no unmeasured intermediate encoder or LLM budgets are interpolated. Horizontal HF base references use zero task-labelled training data and the fixed thinking-off protocol.

These supervised comparisons use additional labelled data and therefore measure a different deployment regime from zero-shot inference. Their strong performance requires the LLM deployment recommendations to be conditional on the availability of training labels. The matched 500-training-label LLM adaptation experiment is reported in Section 5.2; higher-budget baselines do not establish equal-budget superiority. All supervised budgets additionally use the same 2,000 validation labels.

5.2 QLoRA Adaptation with Three Training Seeds

Nine adapter fits completed all twelve preserved tests, giving 216,000 paired base/adapted predictions and 1,800 serial benchmark predictions. Each fit is evaluated separately on the same 3,000 records per condition; Table 3 and Fig. 3 report means and sample standard deviations over three training seeds. Balanced mean weighted-F1 gains range from 3.21 to 9.72 percentage points. Adapted means lie between 0.7833 and 0.7857; these nearby point estimates do not establish model-to-model superiority.

Within this paired test design's 36 model–training-seed–condition–length comparisons, 34 unadjusted 95 % paired F1-change intervals exclude zero in the positive direction. 34 accuracy comparisons pass Holm correction across this family of 36 tests. These within-fit intervals measure uncertainty conditional on saved test outputs; between-fit variation is reported separately. The two intervals containing zero are Gemma seed 2 under reduced imbalance at both character limits; their point gains remain positive, but a reliable improvement is not established in those two comparisons. Overlapping conditions and narrative variants are not independent replications. There are 4 invalid adapted predictions among 108,000 outputs, compared with 73

Table 3 Matched HF base/adapted 500-character results. Each fit uses the same 3,000 test records; entries are mean ± sample standard deviation over three training seeds, not test-population confidence intervals. Training/validation budgets are 500/2,000 records, totalling 2,500 labels.

Model	Test	Base F1	Adapted F1	Gain (pp)	Adapted macro-F1
Gemma 4	A	0.7520 ± 0.0000	0.7841 ± 0.0098	3.21 ± 0.98	0.7841 ± 0.0098
Qwen3	A	0.7099 ± 0.0000	0.7833 ± 0.0092	7.34 ± 0.92	0.7833 ± 0.0092
Phi-4	A	0.6885 ± 0.0002	0.7857 ± 0.0011	9.72 ± 0.10	0.7857 ± 0.0011
Gemma 4	B	0.7730 ± 0.0000	0.7902 ± 0.0076	1.72 ± 0.76	0.7547 ± 0.0046
Qwen3	B	0.7278 ± 0.0000	0.7910 ± 0.0051	6.32 ± 0.51	0.7553 ± 0.0066
Phi-4	B	0.7461 ± 0.0004	0.7917 ± 0.0070	4.55 ± 0.74	0.7570 ± 0.0068

Table 4 Three-fit training and warm serial timing. Means ± sample standard deviations describe three fits. Peak allocated VRAM is the maximum over fits; training time includes validation but excludes loading.

Model	Train/val (min)	Peak (GiB)	Base serial (s)	Adapted serial (s)
Gemma 4	34.74 ± 0.02	7.77	0.482 ± 0.007	0.656 ± 0.008
Qwen3	41.04 ± 0.16	10.04	0.422 ± 0.000	0.539 ± 0.000
Phi-4	40.36 ± 0.28	9.33	0.488 ± 0.002	0.436 ± 0.003

invalid base outputs. Invalid responses remain errors. Supplementary Table S5 reports every fit and condition.

The label-matched classifier comparison uses the same 500 training and 2,000 validation records. The validation-stopped DistilBERT control has balanced weighted F1 0.7150 ± 0.0208; the fixed three-epoch control scores 0.5329 ± 0.1522. TF-IDF/SVM scores 0.7546 at the same training-label budget.

Per-fit adapter–classifier F1 differences use paired cluster bootstrap; accuracy tests correct a separate family of 72 comparisons covering three models, three fit seeds, two conditions, two lengths and two classifiers. On balanced 500-character tests, all nine adapter–SVM F1-change intervals are positive. The corresponding Holm-adjusted accuracy tests pass for all three Phi fits and two fits each for Gemma and Qwen; the remaining Gemma/Qwen comparisons have adjusted p values 0.0768 and 0.0812. All nine comparisons with validation-stopped DistilBERT pass the adjusted accuracy test. These accuracy tests and unadjusted F1 intervals address different estimands and multiplicity. The complete differences and adjusted tests are supplied as reproducibility CSVs; no model-to-model superiority is inferred from point estimates alone.

Supplementary Table S6 reports class F1 over all three fits. Under reduced imbalance, mean adapted credit-card F1 ranges from 0.6398 to 0.6660 while credit-reporting F1 ranges from 0.8481 to 0.8511. Thus the high weighted scores do not remove minority-class weaknesses, including after adaptation.

Table 4 reports training and serial timing. Adapted balanced serial means range from 0.436 to 0.656 s across model averages. These measurements concern unmerged adapters and a fixed 100-record warm serial benchmark per fit, not concurrent throughput. The

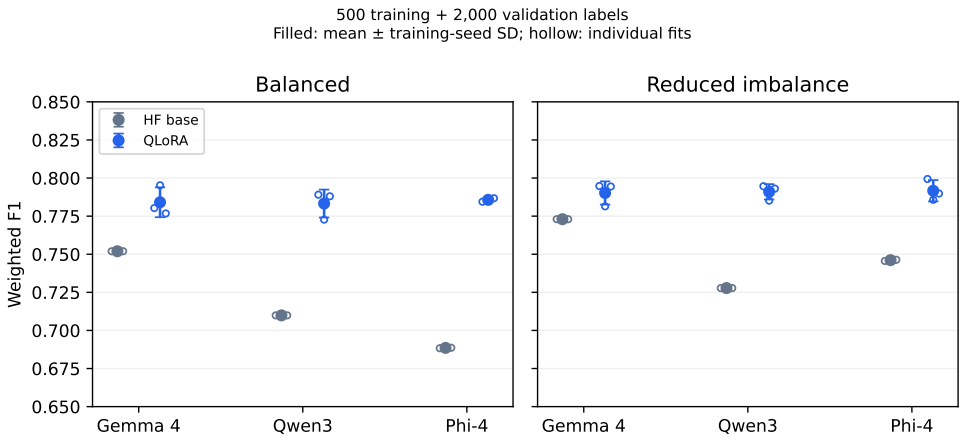


Fig. 3 Matched HF base/adapted mean weighted F1 over three training seeds on identical 3,000-record tests. Error bars show sample standard deviation over fits, not bootstrap confidence intervals. Hollow points show individual fits; the displayed vertical range is 0.65–0.85.

same benchmark records a decrease in Phi-4's mean output length from 13.01 tokens without its adapter to 9.34 with it; Gemma and Qwen remain near 9.5 and 9.3 tokens in both modes. Shorter Phi completions are consistent with its latency reduction, but output length and adapter execution are not independently manipulated, so their timing contributions cannot be isolated from these measurements. Supplementary Section 3.1 reports the per-fit counts. Full-test F1 uses batched outputs; verification scripts compare source hashes, labels, saved responses and artifacts and record serial/batch prediction differences separately.

5.3 Controlled Thinking-On/Off Comparison

We evaluate current pinned Gemma 4 12B and Qwen3 14B GGUF artifacts on the same 500 unique inputs sampled from the first preserved balanced test subset (100 per class, at most 500 characters). An excluded ten-record pilot and a pre-evaluation amendment fix a common 4,096-token output cap, including reasoning. Within each model, only the thinking flag changes: prompts, artifact, temperature zero, context and runtime are fixed. Warm serial requests alternate mode order. Only final-answer JSON is scored; invalid and length-terminated outputs remain errors. Table 5 reports this base-model control; it is separate from HF adaptation. Thinking increases Qwen3's weighted F1 by 5.23 percentage points (unadjusted paired 95 % interval [+1.91, +8.51]); Gemma's change is 1.49 points ([-1.31, +4.19]), leaving its improvement inconclusive. Exact accuracy McNemar tests, Holm-adjusted across the two models, give $p = 0.004372$ and $p = 0.7798$, respectively. These accuracy tests are distinct from the unadjusted F1 intervals.

Class-level point estimates in Supplementary Table S8 increase for all five Qwen3 categories, whereas Gemma's F1 decreases for mortgages/loans (0.7527 to 0.7312) and credit cards (0.7400 to 0.7363), despite its positive overall change. These patterns are descriptive; no class-specific significance tests were performed.

Table 5 Thinking policies on the same 500 balanced inputs. Latency is warm serial HTTP wall time, including full-response transfer and excluding initial loading; it is separate from HF generation timing. F1 is weighted and equals macro-F1 under equal class support. Latency and generated tokens are means; tokens include reasoning. Invalid outputs remain in the denominator.

Model	Thinking	F1	Invalid	Latency (s)	Tokens
Gemma 4	Off	0.7298	0	0.28	13.8
Gemma 4	On	0.7447	14	6.19	642.8
Qwen3	Off	0.6926	0	0.25	9.2
Qwen3	On	0.7449	0	8.43	360.1

Mean latency increases from 0.25 to 8.43 s for Qwen3 and from 0.28 to 6.19 s for Gemma. All 14 invalid Gemma thinking-on outputs (2.8 %) exhaust the shared output budget without a final answer; the other three configurations have none. The comparison therefore estimates thinking-policy performance under a finite total-token budget, conditional on this subset and these artifacts. Prompt and training-seed robustness are outside this comparison. Supplementary Section 3.2 provides the protocol and class scores.

5.4 Zero-Shot GGUF Benchmark: Balanced Evaluation

Table 6 presents the full results for Experiment A, reported as means over three test subsets. Gemma 4 has the highest mean weighted F1 (0.774 ± 0.016), followed by Qwen3 14B (0.750 ± 0.011) and gpt-oss (0.749 ± 0.018). These are descriptive ranks: small subset standard deviations do not establish every pairwise difference. Gemma's unrecorded thinking mode also prevents attributing its result to reasoning being enabled or disabled.

Phi-4 combines F1 0.706 with low recorded latency. Mistral-Nemo has the lowest F1 (0.540 ± 0.025), despite few invalid outputs; its deficit therefore concerns classification quality in the saved configuration rather than primarily output format.

Qwen3 14B, gpt-oss and Qwen3 8B produce no invalid outputs in these balanced tests. Llama 3.1 and Mistral 7B average 66.7 and 58.7 invalid outputs per subset, respectively. Gemma averages 3.7 empty responses; retained logs do not identify their cause. Phi-4 averages one invalid output.

Exploratory exact two-sided McNemar tests compare Gemma with four leading alternatives in each of three test subsets. Treating these as a single family of twelve comparisons gives a Bonferroni threshold of 0.05/12. Gemma's accuracy advantage over DeepSeek-R1 14B and Qwen3 8B passes this threshold in all three subsets. Their largest raw p-values are 0.00296 and 0.000811, respectively. Neither Qwen3 14B nor gpt-oss passes this twelve-test threshold in any subset. The reference model was selected after inspecting scores, so these comparisons remain exploratory and do not establish an F1 ranking or equality between non-significant pairs.

Table 6 Experiment A Results — Balanced ($N = 1{,}000$, mean of 3 test subsets; sample std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
`gemma4:12b`	U	0.773(±.017)	0.774(±.016)	0.774	17.12	3.7
`qwen3:14b`	R	0.751(±.011)	0.750(±.011)	0.750	5.73	0.0
`gpt-oss:20b`	R	0.748(±.020)	0.749(±.018)	0.749	4.13	0.0
`deepseek-r1:14b`	R	0.724(±.006)	0.724(±.007)	0.724	9.33	1.3
`qwen3:8b`	R	0.717(±.003)	0.717(±.002)	0.717	5.19	0.0
`phi4:14b`	S	0.705(±.012)	0.706(±.011)	0.706	0.54	1.0
`deepseek-r1:8b`	R	0.684(±.013)	0.683(±.012)	0.683	5.14	2.0
`gemma2:9b`	S	0.680(±.006)	0.676(±.008)	0.676	0.67	0.3
`mistral:7b`	S	0.593(±.002)	0.587(±.004)	0.587	0.23	58.7
`llama3.1:8b`	S	0.578(±.005)	0.587(±.003)	0.587	2.97	66.7
`mistral-nemo:12b`	S	0.559(±.019)	0.540(±.025)	0.540	0.59	0.7

T: R=Reasoning, S=Standard, U=Unverified mode. OL=Invalid output mean. Lat.=Mean latency.

For Experiment A, macro-F1 is numerically identical to weighted F1 because all five classes have equal support. Per-class inspection reveals that even the best model is uneven across labels: `gemma4:12b` reaches F1 0.825 on `retail_banking` and 0.819 on `mortgages_and_loans` but drops to 0.718 on `debt_collection` and 0.743 on `credit_reporting`. Table 9 reports the leading models' class scores.

Supplementary Fig. S2 shows the balanced F1 distribution. Supplementary Figs. S3, S4, and S5 show confusion matrices for gpt-oss, Phi-4, and DeepSeek-R1 14B, with invalid outputs in a separate error column. Debt collection is frequently predicted as credit reporting in these matrices, even for models with almost no invalid outputs. Llama 3.1's larger invalid-output count primarily reflects refusals rather than invented category names.

5.5 Experiment B: Reduced-Imbalance Evaluation

Table 7 reports three-subset means for Experiment B. All eleven models have numerically higher accuracy and weighted F1 than in A. Gpt-oss leads with F1 0.782 ± 0.016, followed by Gemma (0.779 ± 0.014) and Qwen3 14B (0.772 ± 0.013). Llama and Mistral retain substantial invalid-output counts. Higher aggregate scores under different class proportions do not establish improved robustness.

Macro-F1 exposes the difference from weighted F1: gpt-oss scores 0.740 versus 0.782, and Llama 3.1 scores 0.564 versus 0.647. Class-specific scores in Table 9 show where the majority class masks weaker minority performance.

5.6 Cross-Experiment Comparison

Fig. 4 provides a side-by-side view of accuracy across both experiments. All models improve numerically from Experiment A to Experiment B. `mistral-nemo:12b` has the largest gain (Δacc $= +0.094$), despite few invalid outputs. `mistral:7b` ($+0.055$) and `llama3.1:8b` ($+0.054$) also improve, but these shifts cannot be attributed solely

Table 7 Experiment B Results — Reduced imbalance (N = 1,000, mean of 3 test subsets; sample std in parentheses)

Model	T	Acc.	F1w	F1m	Lat.(s)	OL
`gpt-oss:20b`	R	0.782(±.017)	0.782(±.016)	0.740	4.01	0.0
`gemma4:12b`	U	0.776(±.015)	0.779(±.014)	0.741	10.61	2.7
`qwen3:14b`	R	0.772(±.012)	0.772(±.013)	0.730	5.53	0.0
`qwen3:8b`	R	0.766(±.008)	0.762(±.007)	0.719	6.97	0.3
`phi4:14b`	S	0.753(±.003)	0.754(±.002)	0.696	0.58	0.0
`deepseek-r1:14b`	R	0.746(±.017)	0.748(±.015)	0.700	7.32	0.3
`deepseek-r1:8b`	R	0.738(±.012)	0.731(±.011)	0.677	4.46	2.3
`gemma2:9b`	S	0.720(±.012)	0.719(±.014)	0.656	0.67	0.3
`mistral-nemo:12b`	S	0.653(±.011)	0.641(±.013)	0.538	0.59	0.7
`mistral:7b`	S	0.649(±.010)	0.661(±.010)	0.566	0.23	42.3
`llama3.1:8b`	S	0.632(±.006)	0.647(±.012)	0.564	0.56	57.3

T: R=Reasoning, S=Standard, U=Unverified mode. OL=Invalid output mean. Lat.=Mean latency.

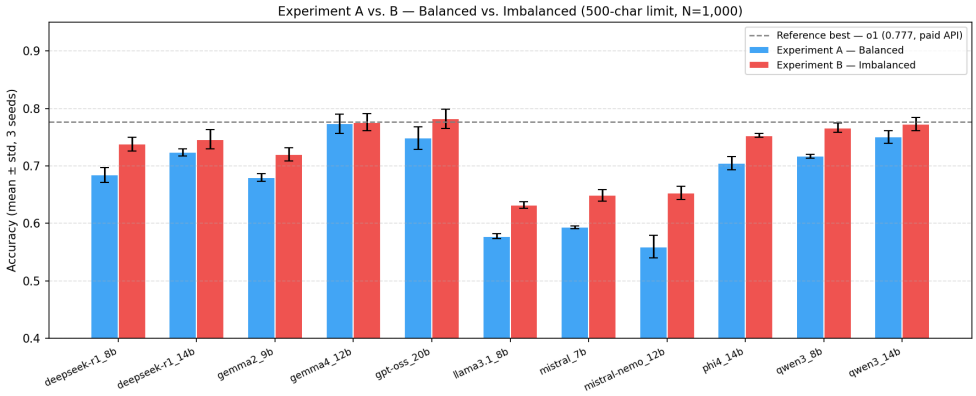


Fig. 4 Accuracy comparison (mean ± sample std over 3 test subsets) across Experiment A (Balanced, blue) and Experiment B (Imbalanced, red) for all models (N=1,000). The dashed reference line marks the best published paid-API result (o1, 0.777) from [8]. The upward shift in Experiment B partially reflects class imbalance and does not isolate improved class-conditional performance.

to invalid outputs. Equal-prior reweighting separates the arithmetic effect of class proportions from class-conditional performance differences. The conditions use partly overlapping but non-identical records, so this does not isolate a causal effect of class imbalance.

5.7 Comparison with Paid-API Models

Table 8 places local balanced results alongside the published API results of Roumeliotis et al. [8]; Supplementary Fig. S6 shows the complete comparison. Gemma's mean F1 0.774 is numerically close to GPT-4.5 (0.7763) and o1 (0.7764). Qwen3 14B and gpt-oss have point estimates above GPT-4o (0.7358) and below Claude 3.7 Sonnet (0.7626).

Table 8 Balanced zero-shot performance and third-party API fees per 1,000 predictions. Commercial scores and reported fees are from [8]; local scores summarise three 1,000-record subsets. Hardware, electricity and maintenance are excluded.

Model	Source	Accuracy	F1	API fee/1k
o1	Paid API	0.777	0.7764	$24.24
GPT-4.5	Paid API	0.774	0.7763	$12.32
`gemma4:12b`	**Open-weight**	**0.773±.017**	**0.774±.016**	**$0.00**
Claude 3.7 Sonnet	Paid API	0.765	0.7626	$0.64
Gemini 2.0 Flash	Paid API	0.758	0.7586	$0.04
`qwen3:14b`	Open-weight	0.751±.011	0.750±.011	$0.00
`gpt-oss:20b`	Open-weight	0.748±.020	0.749±.018	$0.00
GPT-4o	Paid API	0.736	0.7358	$0.45
`deepseek-r1:14b`	Open-weight	0.724±.006	0.724±.007	$0.00
`qwen3:8b`	Open-weight	0.717±.003	0.717±.002	$0.00
`phi4:14b`	Open-weight	0.705±.012	0.706±.011	$0.00

Exact reference test identifiers and predictions are unavailable, so these cross-study observations establish neither paired superiority nor statistical equivalence.

The table compares the model-API fees reported by the reference study: o1 was reported at $24.24 per thousand predictions, while local inference incurs no third-party model-API charge. Hardware, electricity and maintenance are excluded from this comparison.

5.8 Narrative Length Sensitivity

Supplementary Fig. S8 compares mean accuracy on 250-character-limit versus 500-character-limit narratives for the balanced condition (both at N=1,000). The overall ranking is highly stable: no model changes its relative rank by more than one position between the two length limits. Most models improve by 0–3 percentage points moving from shorter to longer narratives, with `deepseek-r1:14b` (+1.6 pp) and `llama3.1:8b` (+2.8 pp) showing gains when additional text from the same complaints is retained.

Pairing all 3,000 balanced predictions and applying Holm correction over 22 model–condition tests gives significant accuracy changes for `llama3.1:8b` ($p_{\text{adj}} < 0.001$) and `qwen3:14b` ($p_{\text{adj}} = 0.0234$). The DeepSeek-R1 14B change does not pass this corrected threshold ($p_{\text{adj}} = 0.0879$). `gpt-oss:20b` and `mistral-nemo:12b` show virtually no change (approximately 0.13 pp and 0 pp, respectively), indicating little observed sensitivity to narrative length in this range. The stability of the top-cluster ranking suggests that the ordering is not driven solely by these two length settings; this does not establish robustness to longer narratives.

5.9 GGUF Request Latency

Tables 6 and 7 report per-request means for the 500-character GGUF tests. In A, Mistral 7B is fastest (0.23 s), while Phi-4 has the highest F1 among configurations

Table 9 Per-Class F1 for Leading Models (Exp. A and B, mean over 3 subsets of N=1,000)

E	Model	Retail	CreditRep.	Card	Mortg.	Debt
A	`gemma4:12b`	0.8247	0.7432	0.7634	0.8194	0.7178
A	`qwen3:14b`	0.7855	0.7212	0.7296	0.8097	0.7019
A	`gpt-oss:20b`	0.7939	0.7137	0.7394	0.7953	0.7045
B	`gpt-oss:20b`	0.7509	0.8527	0.6654	0.7403	0.6919
B	`gemma4:12b`	0.7687	0.8452	0.6713	0.7448	0.6769
B	`qwen3:14b`	0.7353	0.8438	0.6542	0.7426	0.6719

E = Experiment; Mortg. = Mortgages & Loans.

averaging below one second (0.54 s, F1 0.706). Gemma is slowest at 17.1 s; missing thinking and token logs prevent explaining that latency. These benchmark request means are reported separately from the paired HF and thinking-control timings. Supplementary Fig. S7 provides the complete Experiment B F1 plot.

5.10 Per-Class Breakdown for Leading Models

Table 9 reports per-class F1 for the three leading models in each experiment, confirming that aggregate weighted F1 masks important minority-class variation. In Experiment B, the four non-credit-reporting categories have lower support (10.0–18.2 %) than credit reporting (51.1 %); "minority" refers to this support. All three models jump to $\geq$0.84 F1 on `credit_reporting` while minority classes remain in the 0.65–0.77 range, showing the class-specific variation discussed in Section 6.

5.11 Additional Statistical and Output Audit

We estimate 95 % percentile confidence intervals using 5,000 bootstrap resamples of the pooled 3,000 predictions in each 500-character condition. Resampling is stratified by true class and clusters identical normalised narratives, so repeated texts receive the same bootstrap weight. These intervals condition on the saved test sets and model outputs, and do not include uncertainty about pre-training, prompt choice, or commercial reference results. The pooled estimators differ slightly from means of subset-specific F1. Table 10 reports selected balanced-test estimates.

The published GPT-4o point estimate (0.7358) falls within the local Qwen3 14B and gpt-oss intervals. This observation does not constitute a paired comparison or equivalence test, because reference predictions and sampling uncertainty are unavailable.

To examine class-prior effects, we rescale each row of the pooled Experiment B confusion matrix to equal support, retaining invalid predictions as an additional error column, then recompute metrics. Fractional counts are retained without rounding; this reweights observed errors rather than improving predictions. Supplementary Section 2 gives the transformation. For `mistral-nemo:12b`, observed accuracy 0.6530 becomes 0.5562, close to Experiment A's 0.5593. For `gpt-oss:20b`, accuracy 0.7820 becomes

Table 10 Pooled balanced weighted F1 with 95 % cluster-bootstrap confidence intervals ($N = 3{,}000$).

Model	Pooled F1	95 % interval
`gemma4:12b`	0.7738	[0.7586, 0.7882]
`qwen3:14b`	0.7498	[0.7344, 0.7650]
`gpt-oss:20b`	0.7497	[0.7344, 0.7652]
`phi4:14b`	0.7057	[0.6892, 0.7216]

0.7480, close to Experiment A's 0.7483. The weighted F1 values similarly change from 0.6413 to 0.5326 and from 0.7825 to 0.7495, respectively. Thus most of these aggregate gains can be explained arithmetically by different class proportions; this is not a causal experiment with the class prior independently manipulated.

Reparsing all 132 saved response files with a JSON decoder, accepting only an explicit string-valued `product` field, changes no in-taxonomy prediction in the recorded GGUF benchmark outputs. Across all Gemma 4 files, 37 saved responses are empty; their causes cannot be distinguished without separate thinking and termination logs. Representative Llama parse errors are explicit refusals, whereas Mistral 7B often returns parseable labels such as `online_banking` or `not_applicable` outside the taxonomy. These are distinct deployment failures even though both count as errors.

6 Discussion

6.1 Thinking, Reliability and Serving Cost

The controlled experiment in Section 5.3 shows that thinking has model-dependent value under the same total output budget. Qwen3 gains 5.23 weighted-F1 percentage points, with a paired interval excluding zero; Gemma's 1.49-point change is inconclusive. Mean latency rises approximately 34-fold and 22-fold, respectively. Gemma also exhausts the output budget on 2.8 % of thinking-on requests. Enabling thinking therefore requires evaluating both final answer quality and completion reliability. This paired control isolates the thinking-policy comparison within each fixed model configuration.

Invalid outputs require diagnosis beyond a single failure count. GGUF benchmark Llama failures often contain refusals, whereas Mistral 7B often emits valid JSON with unsupported labels. Format constraints alone would not enforce a correct classification or remove refusals; their effect was not tested here. Within-family score differences also mix size, generation, training recipe or backbone, as detailed in the supplementary material.

6.2 Label Budgets and Deployment Choices

At the matched 2,500-label budget (500 training and 2,000 validation), adapted LLM mean weighted F1 exceeds the SVM point estimate by approximately 2.9–3.1 percentage points on the balanced tests. This gain accompanies warm serial latency of 436–656 ms for the GPU-served LLMs, compared with 0.29 ms for CPU SVM on the

same workstation. The observed trade-off makes response-time requirements central to model selection. These point estimates do not establish model-to-model superiority.

With 2,000 training labels plus the same validation budget, SVM reaches 0.7908; the 20,000-record classifiers score still higher. These observed regimes support direct classifiers when suitable labels are available. The unmeasured LLM learning curve prevents identifying an equal-budget crossover.

Among the GGUF zero-shot configurations, Gemma offers the highest balanced point estimate, Phi-4 the highest F1 among configurations averaging below one second, and Qwen3 8B a smaller artifact with few invalid outputs. These are conditional trade-offs, not deployment guarantees: latency depends on hardware and runtime settings, artifact size does not establish runtime memory, and serial means do not establish tail latency or concurrent throughput.

Local inference avoids third-party model-API fees and keeps new complaint text on local infrastructure. The cross-study commercial comparison supplies context rather than proof of equal quality or total cost savings. Macro-F1 and class scores remain necessary under imbalance: equal-prior reweighting explains much of Experiment B's aggregate gain, and adaptation retains minority-class weaknesses.

6.3 Limitations

Data and generalisation. The 500-character filter and normalisation change the evaluated population and may remove useful cues. The paired 250/500-character tests address truncation within short complaints, not performance on naturally longer narratives. Partly overlapping test conditions are not independent replications. English narratives, the five-category taxonomy and single instruction template further limit generalisation; pre-training contamination cannot be ruled out.

Model and training scope. The eleven-model benchmark lacks complete artifact digests and thinking logs; differing quantisation, architectures and training recipes confound broad family comparisons. The thinking control pins current artifacts but uses one fixed 500-record subset and a finite output budget. LLM adaptation uses only 500 training records and three seeds on fixed training/validation records; LoRA rank and target-module choices are not ablated. Within-fit test intervals and between-fit standard deviations do not establish robustness across new training corpora, prompts or larger label budgets. Every supervised budget also requires 2,000 validation labels.

Operational scope. HF serial timing uses 100 fixed balanced examples after warmup and excludes output decoding and network serving; thinking-control timing includes the full HTTP response. They are not interchangeable timing boundaries. Full-test HF scores use batched outputs, which differ from serial predictions in 11 of 1,800 benchmark cases. Concurrent serving and total ownership cost were not measured. Zero model-API expenditure does not imply free local computation. Abstention, human escalation and asymmetric misrouting costs are outside the evaluated routing task.

7 Conclusion

We compared local zero-shot configurations, three-seed QLoRA adaptation, direct classifiers and controlled thinking policies on preserved CFPB tests. At 500 training

plus 2,000 validation labels, adapted balanced weighted F1 averages 0.7833–0.7857, improving matched bases by 3.21–9.72 percentage points. These means exceed the label-matched SVM point estimate, with higher serial serving latency. Direct classifiers improve at larger observed budgets; LLM adaptation beyond 500 records was not evaluated, so the equal-budget crossover remains unknown.

Thinking improves Qwen3 under the controlled budget, but Gemma's change is inconclusive and both incur substantial latency increases. The practical choice depends on label availability, completion reliability and serving requirements. Minority-class performance requires separate inspection even after adaptation. Findings remain conditional on this short-narrative corpus, fixed prompts and measured serial settings; longer narratives, temporal shift, concurrent throughput, tail latency and total ownership cost remain unevaluated.

Declarations

Funding. This research received no external funding.

Conflict of interest. The authors declare that they have no conflict of interest.

Ethics approval and consent to participate. Not applicable: analysis of an existing public corpus involves no participant recruitment or intervention. Public narratives may retain identifying or financial information; local processing alone does not ensure privacy.

Consent for publication. Not applicable.

Data availability. The study uses the publicly available CFPB Consumer Complaints dataset published on Kaggle [28].

Materials availability. Not applicable.

Code availability. The original benchmark repository is available at: https://github.com/muhammettan28/ConsumerComplaintwithLLMs The adaptation pipeline records pinned model revisions, split manifests, source/adapter hashes and generated run artifacts for reproducibility.

Author contributions. Muhammet Tan and Halit Bakır contributed to the study conception, experimental design, implementation, analysis, and manuscript preparation.

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